Option Pricing with Monte Carlo

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Abstract

The Black-Scholes Model was the first widely used method to calculate the theoretical value of an option contract. It derives the price of a European-style call option from an assumption of randomness, as seen in Brownian Motion. I studied the model under the guidance of a graduate student, using probability theory, stochastic calculus, and differential equations.
I then used Python to compare actual (Black-Scholes) and derived (Monte-Carlo) option prices, and plotted graphs to visualize the convergence of Monte-Carlo prices to the actual price and how option prices respond to different variables.

Black-Scholes Equation

We assume that the stock price follows the Stochastic DE:

$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t$$

By Ito's Lemma:

$$df = \left(\frac{\partial f}{\partial S}\mu S + \frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S^2\right) dt + \frac{\partial f}{\partial S}\sigma S \, dW_t$$

For the portfolio Π of -1 option and $\partial f / \partial S$ shares:

$$\Delta\Pi = -\Delta f + \frac{\partial f}{\partial S}\Delta S$$

This portfolio is riskless, so it must earn the risk-free rate:

$$\Delta\Pi = r\Pi\Delta t$$

Combining the two:

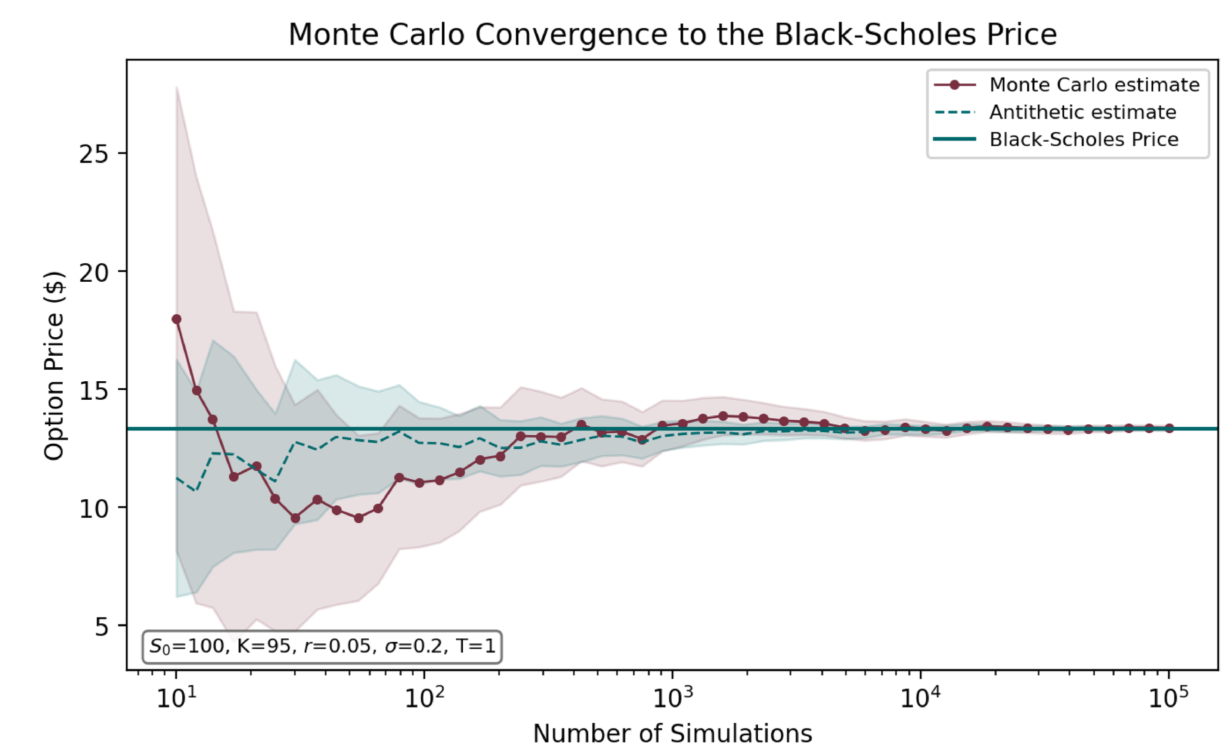
$$-\left[\frac{\partial f}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2}\right]\Delta t = r\left[-f + \frac{\partial f}{\partial S}S\right]\Delta t$$

which implies the Black-Scholes partial differential equation:

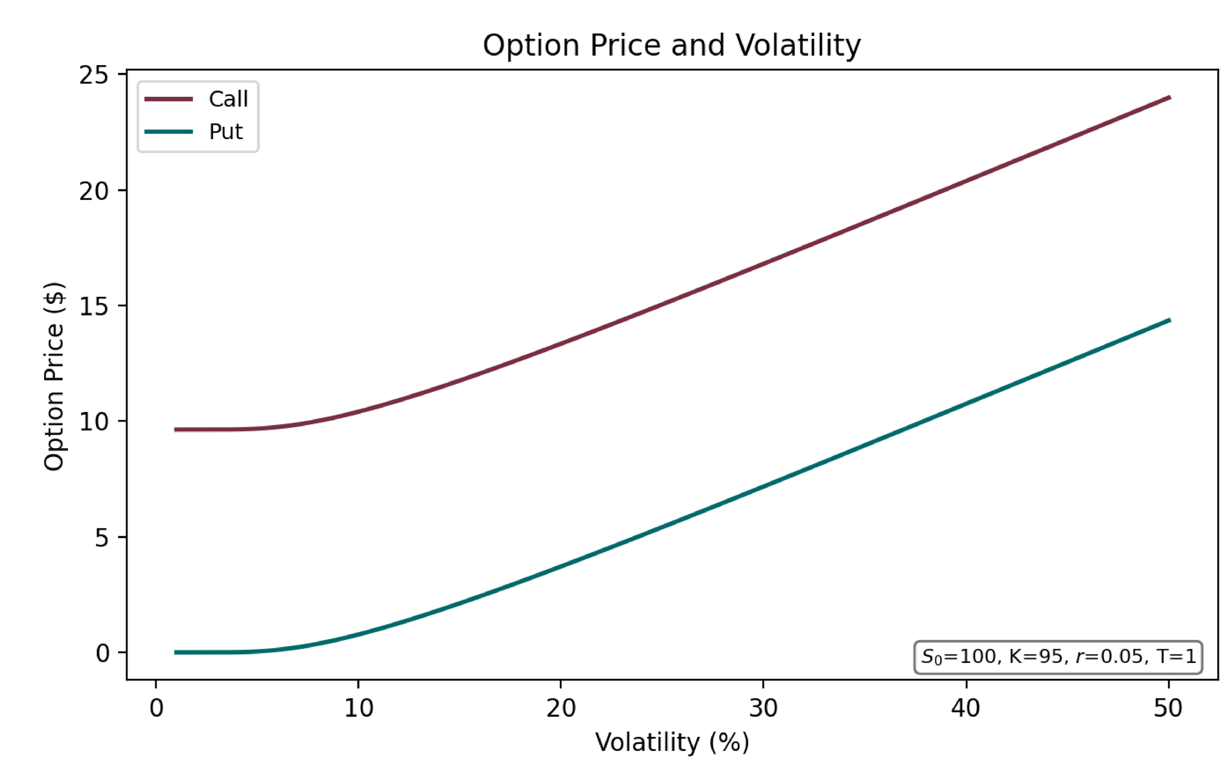
$$\frac{\partial f}{\partial t} + rS\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} = rf$$

where S is the current stock price, K the strike price, f the price of an option depending on S and t, t the time to expiration, r the risk-free rate of interest, σ the rate of volatility, and Π the value of the portfolio.

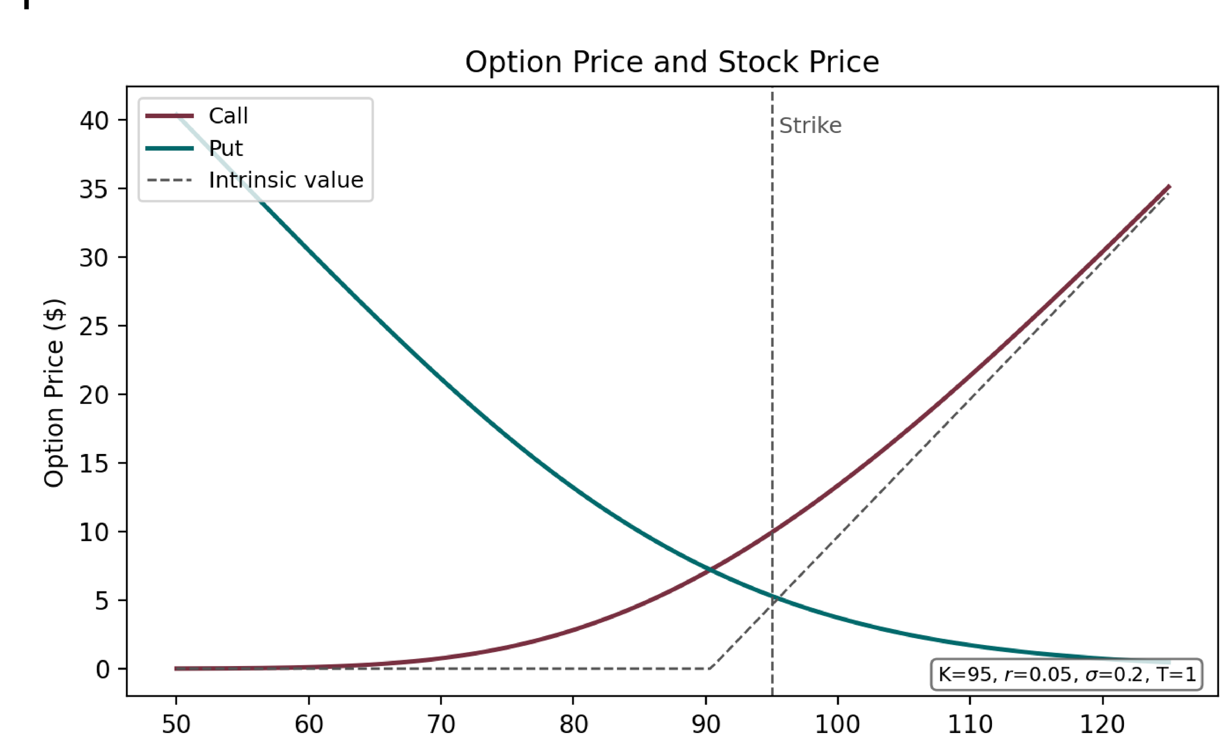
Graphs



Monte-Carlo estimates converge to the Black-Scholes price as simulations grow; antithetic sampling gives a tighter confidence band.



Higher volatility raises option prices: more volatility means more upside potential.



The call price rises with the stock price above the strike and flattens toward zero below it.

Modelling & Motivations

The Black-Scholes PDE can be solved to give the call price C and put price P:

$$C = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

$$P = Ke^{-rT} N(-d_2) - S_0 N(-d_1)$$

A call option gives the holder a right, but not an obligation, to buy a stock at a predetermined price on a later date; a put option gives the right to sell. Options give exposure to a stock with a smaller investment, but are generally much riskier.
These models let traders control the risks of options trading and enabled the development of the derivatives market. Their strong assumptions do not always hold, so prices will not always match the market, but they still convey real information about how options behave.
I performed the calculations and plotted the graphs in Python using the NumPy, SciPy, and Matplotlib libraries.

Question

What do you think the option price will be with these variables?

Current stock price	\$100
Strike price	\$95
Time to expiration	1 year
Risk-free interest rate	5%
Volatility	20%

Conclusion

Adjusting different variables in the Black-Scholes Model and Monte Carlo calculations results in a deeper understanding of how option prices vary and has important real world implications.
I would like to thank Pervez Ali for being a wonderful mentor for this project.